

COMP1047 AY2526: Final Exam Review Guideline

- **NO** calculator is allowed.
- We will provide you the MIPS Ref Card appended at the end of the exam paper.
- Dictionary is permitted as per usual exam rules.
- The final exam covers the three (3) aspects that have been taught in the lecture and exercised in the labs:
 - Computer Networking
 - Computer Architecture Basics and MIPS programming
 - ISA and CPU Design

Computer Networks

- Week 6: Introduction to Computer Networks
 - Data communication components and data flow
 - General concepts about networking
 - Network topologies
 - Type of networks (geographical, etc.)
 - What's a protocol
 - Overview of the Internet: network edge (end systems), access networks
 - Network core functionalities (forwarding and routing)
 - Packet-switching
 - Internet structure
- Weeks 7: The Internet: TCP/IP and OSI models. The Application layer
 - Network performance: packet delays, loss and throughput
 - Protocol layering
 - TCP/IP model
 - OSI model
 - Encapsulation
 - Protocols at each layer
 - History of the Internet
 - Principles of network applications (client-server and P2P paradigms, processes and sockets, etc.)
 - Web and HTTP
 - The Domain Name System (DNS)

Computer Networks

- Week 8: Core Internet Protocols 1: Transport Layer
 - Email, SMTP, IMAP (belongs to previous week)
 - Transport layer services (communication between processes)
 - Principles behind transport layer: multiplexing, reliable data transfer, flow control, congestion control
 - User Datagram Protocol
 - Transmission Control Protocol
 - Evolution of transport-layer functionality
- Weeks 9: Core Internet Protocols 2: Network Layer
 - Network layer overview
 - What's inside a router (I/O ports and switching fabric)
 - The Internet Protocol (IP)
 - Addressing (IPv4) & subnetting
 - IPv6
 - NAT
 - DHCP
 - Generalized Forwarding
 - Middleboxes
 - The 'control plane'
 - Routing protocols
 - Software Define Networking
 - Internet Control Message Protocol

Computer Networks

- Week 10: Link Layer & Network Security Brief
 - Link layer concepts (services, interfaces, etc.)
 - MAC addresses
 - Address Resolution Protocol
 - Ethernet Switch concepts (self-learning, etc.)
 - Switches vs. routers
 - A day in the life of a web request (all together!)
 - What is network security?
 - Cryptography
 - Securing TCP connections: TLS
 - Operational security: firewalls and IDS
 - Port scanning

Computer Networks

- Exam:
 - 26 May 2026
 - 9.30-11.30
 - YANG Fujia Building-118 & YANG Fujia Building-226 (provisional)
- 2 hours
- 2 parts: 6 questions
 - CPU: 3 questions (50 marks)
 - Networks: 3 questions (50 marks)
- Past papers in Moodle

Computer Networks

Exam Preparation:

- General recommendation:
 - Study...
 - ...Everything...
 - ...(Very) Well....
 -Focus on Understanding 😊
- Everything is in the slides, take it easy and be systematic. Avoid cramming!
- “Lección dormida, lección aprendida”
 - “A lesson slept is a lesson learned” 😊
- Practice
 - Past papers in Moodle – no solution provided
 - Same content but different focus and labs for the networks part

Last Words

- Thank you for attending! (to those who attended)
- Thank you for participating! (to those who participated)
- It was truly a pleasure to have shared path with you. I hope to engage with you in the future!
- Hopefully, in the Computer Security module 😊

Best of luck on your exams!